Remote Sensing Image Semantic Segmentation Utilizing Geographical Element Association Features

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***Abstract*—Semantic segmentation of high-resolution remote sensing images remains challenging due to complex spatial structures, fine-grained category variations, and hierarchical semantic dependencies. Existing deep learning models often treat semantic categories as independent, flat labels, which leads to inconsistent predictions across hierarchical levels (e.g., a pixel predicted as tree but not as vegetation) and weak generalization in heterogeneous landscapes. To address these issues, we propose hierarchical attention gate multiresidual UNet (HAG Net), a novel framework that explicitly models hierarchical semantics and enforces cross-level consistency. First, a multiresidual encoder–decoder backbone with hierarchical attention gates (HAGs) enhances multiscale representation while filtering irrelevant background noise. Second, a mixture-of-head (MoH) attention module enables bidirectional semantic interaction between coarse- and fine-level features, mitigating error propagation caused by unidirectional designs. Third, a hierarchical interaction (HI) loss introduces a dynamic category interaction matrix to adaptively constrain predictions, ensuring consistency across levels. Extensive experiments on two large-scale datasets, GID (5- and 15-class) and Ascend Cup 2020 (8- and 17-class), demonstrate that HAG Net consistently outperforms state-of-the-art methods, including DeepLabv3+, CGGLNet, and MAE-BG. Specifically, HAG Net achieves up to 4.5% improvement in FWIoU and significantly enhances per-class accuracy for small and structurally complex objects. These results confirm the effectiveness of incorporating hierarchical semantics into segmentation networks and highlight the potential of HAG Net for large-scale land-cover mapping, ecological monitoring, and urban planning applications. The source code is available at https://github.com/yangruiqi-kiki/HAG**

***Index Terms*—Deep learning, geographic element modeling, hierarchical semantic segmentation, interaction loss.**

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I. Introduction

As a pivotal technology in remote sensing image interpretation, semantic segmentation provides robust support for the in-depth analysis of high-resolution remote sensing images through pixel-level fine classification [1], and it has been widely applied to land-use classification [2], in geographic information retrieval [3], and in disaster monitoring and prevention [4].

Geographic entities in remote sensing images exhibit complex spatial structures and hierarchical characteristics. The interrelationships among these entities not only influence the spatial distribution of visual data but are also key to scene understanding and interpretation. In high-resolution remote sensing scenarios, different categories of geographic entities often demonstrate significant semantic dependencies and hierarchical relationships, as illustrated in Fig. 1. Effectively capturing and modeling these complex semantic dependencies remains a challenging task. Therefore, developing a segmentation framework capable of effectively capturing semantic relationships and spatial interactions among geographic entities is of paramount importance. Such a framework must comprehensively represent scene information while significantly enhancing the model's generalization performance in multicategory, multiscale environments [5].

To address these challenges, researchers have traditionally relied on various approaches in remote sensing image analysis. Classical methods such as conditional random fields (CRFs) [6], [7] and Markov random fields (MRFs) [8], [9] have played important roles in modeling spatial relationships and semantic dependencies. However, these approaches rely on handcrafted features and predefined model assumptions, which constrain their ability to capture complex, nonlinear semantic relationships in diverse geographic scenes.

The rapid progress of deep learning has driven substantive advances in semantic segmentation. Model families represented by fully convolutional networks (FCNs) [10], augmented with attention mechanisms [11], [12], [13] and

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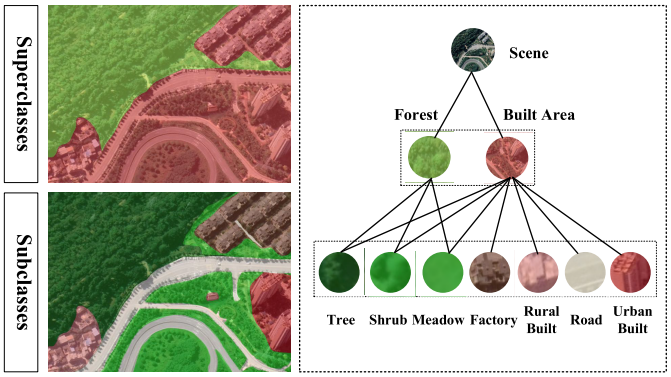


Fig. 1. Multilevel and complex relationships between superclass and subclass of geographic elements. Subclasses are not necessarily restricted to conventional classifications and can be associated with multiple superclasses. In urban scenes, although "tree" is sometimes considered a subclass of "built area," it is not a typical category of buildings. Additionally, as a subclass, "tree" can belong to multiple superclasses, such as "forest" and "built area."

encoder–decoder architectures [14], [15], enlarge the effective receptive field, fuse multiscale context, and strengthen long-range dependency modeling, yielding better interpretation in complex scenes. Compared with traditional methods, deep learning-based models achieve higher recognition accuracy and stronger robustness on remote sensing imagery [16].

However, mainstream methods still have inherent limitations. Most models use an independent category modeling approach. They treat each category as an isolated entity in pixel-wise predictions and rely only on single-category features. This ignores semantic relationships and spatial interactions between categories, reducing effectiveness in complex scenarios. In real-world environments, geographic objects are often spatially coupled and have intricate semantic dependencies. The boundaries between categories are rarely independent. Instead, they are interconnected and often follow a hierarchical structure. Relying only on single-category information increases misclassification rates. It can also disrupt semantic consistency, making models less adaptable to complex environments [17].

In typical urban scenarios, there exist intricate spatial associations and functional dependencies among geographic elements such as buildings, rivers, and forests. Certain subclasses, such as transportation land, are often embedded within dominant regions of other types, such as urban construction land, farmland, or forest areas. Similarly, vegetation elements not only form natural geographic units like forests but also serve as artificial landscape components distributed across residential areas, reflecting the multiattribute and interconnected characteristics of geographic elements across different semantic hierarchies, as illustrated in Fig. 1.

Relying only on single-category information increases misclassification rates. It can also disrupt semantic consistency, making models less adaptable to complex environments. To overcome the limitations of independent category modeling, recent studies have incorporated relational modeling into deep learning frameworks. This improves information propagation and promotes feature sharing across categories. Nonlocal networks and graph convolutional networks (GCNs) have been used in semantic segmentation to capture long-range dependencies and model global context [18], [19]. These methods utilize global feature aggregation and graph-based learning to strengthen intercategory dependency modeling. Yet, two gaps remain: 1) hierarchical information is undermodeled, making nested relations across scales difficult to represent and 2) information propagation remains constrained—unidirectional designs are common, with top-down hierarchical modeling where coarse semantics guide finer predictions (e.g., GLC_FCS10 first partitions major categories and then refines granularity [20]; HSSN constrains pixel labels to follow a "root-to-leaf" path to ensure general-to-specific consistency [21]), and bottom-up feature aggregation where low-level cues are progressively integrated upward (e.g., HPS-Net extracts stage-wise hidden variables to guide main-path selection [5], MFE-Net uses a feature aggregation–diffusion pyramid for bottom-up aggregation [22]. As a result, they fail to capture bidirectional interactions between categories. These limitations make it challenging for current semantic segmentation models to accurately represent complex semantic relationships, especially in multiscale and multilevel geographic objects in remote sensing imagery.

To simulate hierarchical visual perception and effectively integrate multigranularity semantic relationships, we propose a deep hierarchical semantic segmentation method for remote sensing images. The proposed network enables bidirectional interaction at both the architecture and loss levels. Implicit interaction is realized by architectural message passing modules that exchange multiscale features across branches, while explicit interaction is enforced by a learnable category interaction matrix in the objective to directly regulate coarse and fine semantic dependencies. The main contributions are stated as follows.

1) This study proposes a hierarchical association-based deep semantic segmentation framework (HAG Net) to address the limitations of traditional methods in effectively modeling the hierarchical semantic structures of geographic elements. The proposed framework incorporates hierarchical association modeling, establishing multiscale semantic representations that span from regional structural patterns to local surface morphology. This enables the model to simultaneously capture the spatial distribution patterns and hierarchical semantic relationships of geographic elements.
2) Furthermore, this study introduces the hierarchical interaction loss (HI Loss), which leverages a category interaction matrix to reinforce semantic associations between categories, enabling information sharing and collaborative optimization across classes. This mechanism overcomes the limitations of traditional flat structural representations, allowing the segmentation model to maintain higher accuracy and stability in complex backgrounds and multilevel category relationship scenarios.

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II. Related Work

A. Deep Learning for Semantic Segmentation

Semantic segmentation of remote sensing imagery identifies the semantic category of every pixel, enabling automatic delineation and interpretation of ground objects and regions. Traditional approaches struggle with high-resolution data because they cannot fully capture complex spatial structures and contextual cues. Convolutional neural networks (CNNs) [23] overcome this limitation through strong spatial feature extraction and context modeling, greatly improving segmentation accuracy.

Classic architectures such as U-Net [24], DeepLab [25], and PSPNet [26] are widely used to capture multiscale spatial features and lift segmentation performance. The encoder–decoder frameworks, atrous convolutions, and pyramid pooling modules efficiently aggregate contextual information, delivering precise classification across diverse land-cover types. Attention mechanisms [27], [28] and Transformer-based designs push performance even further. By modeling long-range dependencies and intricate spatial patterns, models like Swin Transformer [29], SegFormer [30], and the hybrid DynamicVis [31] achieve state-of-the-art results on many remote sensing benchmarks.

Despite these advances, most methods still rely on flat semantic labels and treat classes as independent. They cannot explicitly model hierarchical or nested relationships—such as vegetation within residential areas or roads between buildings—so their ability to represent and reason over multiscale geographic entities remains limited.

B. Traditional Hierarchical Semantic Segmentation

Traditional hierarchical semantic segmentation methods primarily include CRFs and MRFs [8], [9], [32], which model class co-occurrence dependencies to constrain pixel distributions within target boundary regions, thereby improving classification accuracy. Wang et al. [8] refined the initial classification by incorporating local spectral dissimilarity while integrating class co-occurrence relationships to enhance spatial coherence. Zheng et al. [9] proposed a hierarchical self-learning knowledge inference model (HSKIM) based on the MRF framework, introducing an object-based MRF model to facilitate semantic inference between intralayer and interlayer units. Fu et al. [32] introduced the object-based MRF with hierarchical semantics (OMRF-HSs), enabling effective hierarchical semantic representation in remote sensing imagery and improving the understanding of complex scenes. However, their reliance on predefined classification rules and handcrafted local features constrains their ability to adapt to the diverse semantic patterns and multiscale nested structures commonly encountered in remote sensing scenarios.

Another major direction has focused on explicitly constructing a hierarchical structure during the segmentation process. One approach involves cascaded classifiers, where a sequence of classifiers is trained to iteratively refine predictions. The Auto-Context model proposed by Tu [33], for example, leverages the probability maps from a previous stage as rich contextual input for the next. This creates a sequential, top-down refinement process. An alternative, bottom-up strategy is found in hierarchical region merging techniques, such as region trees and HSeg trees [34], [35]. These methods build a hierarchy by recursively merging adjacent regions based on low-level visual cues like color, texture, or edges. While Xu et al. [34] focused on automating the tree construction, others like Zhang et al. [35] combined it with active learning for supervised pruning. Although both cascaded and merging strategies attempt to represent hierarchical information, their mechanisms are typically governed by unidirectional information flow (either top-down or bottom-up) and depend heavily on heuristic rules rather than adaptive, data-driven optimization.

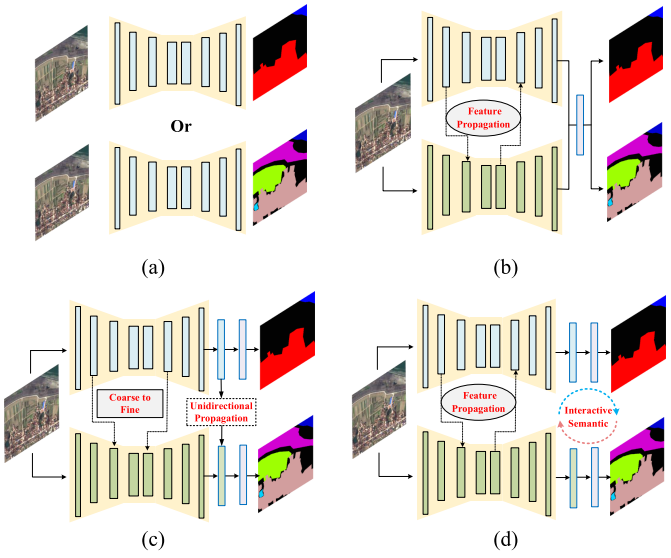


Fig. 2. Comparison of different semantic structure modeling paradigms. (a) Flattening structure performs a single-level segmentation task without semantic hierarchy. (b) Hierarchical feature structure supports multilevel feature processing but does not distinguish semantic granularity at the task level. (c) Multilevel semantic structure considers coarse-to-fine semantics but relies on unidirectional propagation, which may lead to error accumulation. (d) Proposed HAG net incorporates structural feature interaction and cross-level semantic interaction to enhance consistency and collaborative understanding.

Consequently, while these traditional methods introduced valuable concepts for contextual and structural modeling, they share common limitations. Their dependence on handcrafted features, rigid structural assumptions, or nonadaptive heuristic rules makes it challenging for them to generalize effectively to the intricate category relationships and visual diversity present in complex geographic environments. This highlights the necessity for learning-based models that can end-to-end capture and reason about hierarchical semantics directly from data.

C. Interactive Hierarchical Semantic Segmentation

In the deep learning framework, hierarchical semantic segmentation confronts fundamental challenges in modeling taxonomic relationships while maintaining effective cross-scale information propagation. Traditional flat segmentation

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architectures exhibit critical deficiencies in capturing the inherent hierarchical taxonomies of geographic phenomena, as illustrated in Fig. 2(a). These methods fail to adequately represent the nested categorical structures and semantic interdependencies that characterize complex Earth observation scenarios, particularly when dealing with high-resolution imagery containing heterogeneous land-cover patterns.

To alleviate insufficient modeling of semantic dependencies, recent work integrates graph-based relational reasoning and attention-driven intercategory modeling into hierarchically layered network architectures [36], [37], [38], [39], [40]. These architectures enhance semantic coherence across the hierarchy via learnable association matrices and cross-level feature exchange; representative implementations typically combine parent–child pathways with cross-layer feature-consistency constraints. However, such approaches still rely primarily on implicit semantic relationship learning emerging from shared representations, rather than explicit modeling of intercategory interaction patterns. This indirect paradigm struggles to fully represent complex hierarchical dependencies and to capture the fine-grained semantic associations required for accurate geographic entity classification (see Fig. 2(b) for an example).

Semantic hierarchy-based analytical algorithms have been developed based on recent advancements, typically achieving segmentation through coarse-to-fine semantic levels [41], [42], [43], [44] as shown in Fig. 2(c). Jing et al. [41] proposed a recursive coarse-to-fine semantic segmentation framework using image-level category labels. Fu [43]introduced Bayesian techniques into CNN-based hierarchical classification and designed the coarse-to-fine CNN structure. Building on these advances, RSPrompter [42] explicitly integrates semantic category information to generate prompts, coupling foundation "segment-anything" models with prior knowledge to alleviate the reliance on coarse-grained masks and improve the delineation of instances and fine-grained regions. Shenaj et al. [44] proposed a coarse-to-fine knowledge distillation constraint to transfer network capabilities from coarser to finer category sets.

Current hierarchical semantic segmentation methods predominantly rely on unidirectional information propagation, employing either top-down or bottom-up strategies. This architectural limitation presents two fundamental challenges. First, classification errors propagate without effective correction mechanisms, leading to error accumulation within the hierarchical structure and degraded segmentation performance. Second, these approaches enforce strict separation of category features during pixel prediction, failing to capture the inherent structural complexity of visual scenes.

Recent advances have attempted to introduce interlevel interactions, yet they maintain idealized assumptions that subclasses are rigidly confined to their designated parent categories [45]. For instance, "trees" are exclusively classified under "vegetation," disregarding their contextual occurrence in urban environments. This rigid taxonomic constraint inadequately represents the spatial diversity and semantic flexibility inherent in real-world scenarios [21], where subclasses naturally exhibit membership across multiple parent categories.

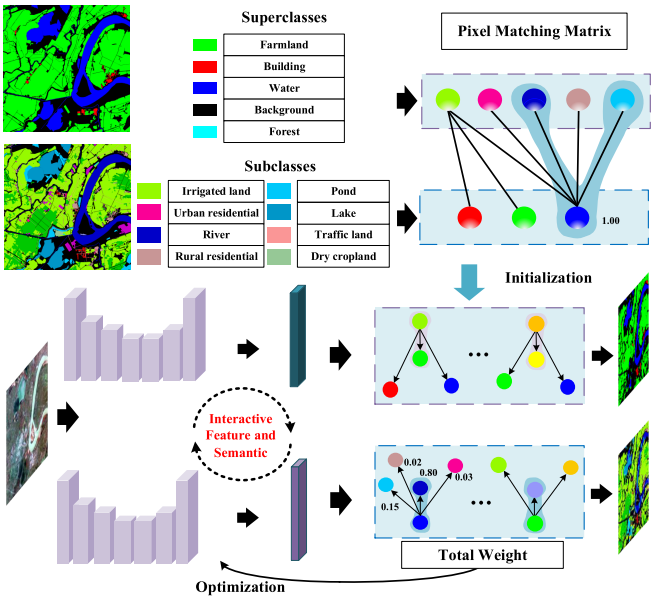


Fig. 3. Overview of the HAG net framework. The model leverages transfer matrices derived from the training set to guide the HI Loss. This loss interactively updates weights between superclass (e.g., water) and subclass (e.g., lake and pond) predictions at the same pixel location to enforce hierarchical consistency.

In contrast to these constraint-rigid methodologies, our approach introduces a soft-consistency framework that facilitates flexible taxonomic associations while accommodating the semantic ambiguity inherent in hierarchical classification problems [Fig. 2(d)]. This framework allows for multilevel category membership and avoids mandatory parent–child constraints, thereby overcoming a fundamental limitation of existing approaches. Through adaptive semantic modeling, our method systematically addresses deficiencies in cross-scale coupling and effectively suppresses the propagation of classification errors through the hierarchical structure.

III. Methodology

Hierarchical semantic segmentation is formulated as a structured multilabel classification problem at the pixel level (see Fig. 3), wherein each spatial location receives semantic annotations across multiple hierarchical levels. Unlike conventional top-down or bottom-up one-way propagation, our network performs query–key/value retrieval across levels: queries at each level attend to the corresponding key–value (K/V) pairs to realize implicit parent–child interactions, which are then dynamically fused with backbone features. Beyond in-network interactions, we encode coarse–fine relations explicitly in the loss via a directed acyclic graph. A category interaction matrix M specifies interclass links and is dynamically injected into the loss to enforce multilabel consistency. This design establishes bidirectional ties between super and subclasses, enables cross-level information sharing and multigranularity integration, and strengthens the modeling of complex hierarchical relations.

To realize these design principles, we propose a hierarchical semantic segmentation network (Section III-A) and an HI

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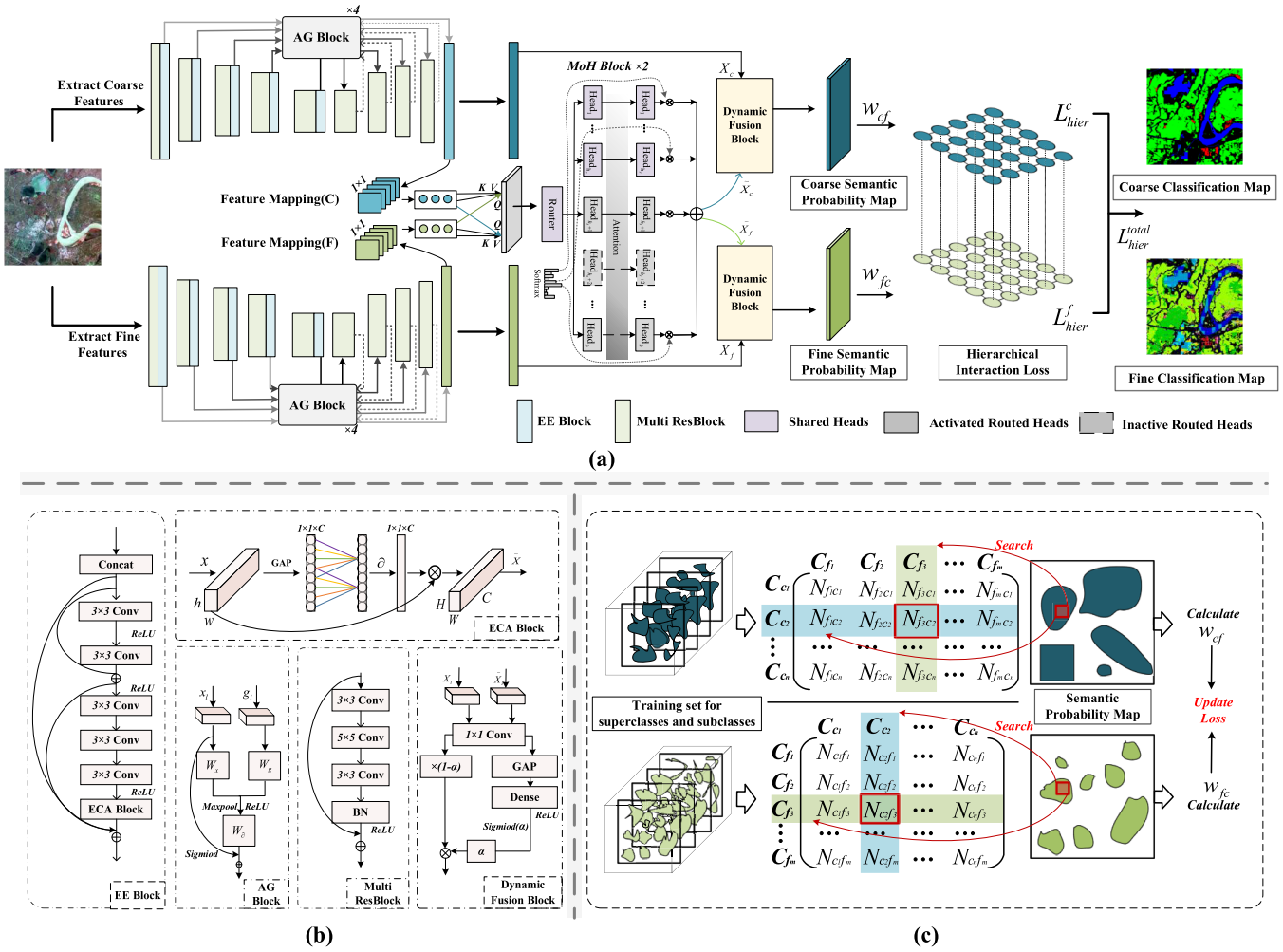


Fig. 4. Proposed HAG net architecture, which consists of encoding and decoding. The encoding consists of four EE blocks and multiple ResBlocks. The decoding consists of four multiple ResBlocks. The skip connection is AG block. The loss function is HI Loss (F represents subclasses and C represents classes). (a) HAG Net. (b) The Components of HAG Net. (c) Hierarchical Interaction Loss.

Loss (Section III-B), with the complete prediction framework presented in Section III-C.

A. HAG Multiresidual UNet

Hierarchical geographic element recognition necessitates adaptive feature representations across diverse semantic granularities, where fine-grained categories require high-resolution spatial details and coarse-grained classes demand broader contextual abstractions. To achieve effective multigranularity feature modeling, we construct a hierarchical attention-guided network architecture that simultaneously captures fine-scale geometric details and coarse-scale semantic contexts.

The structure proposed in this article as shown in Fig. 4 is primarily based on the ResUNet network. The encoder consists of four edge enhancement modules (EE block) and multiscale residual blocks (Multi-ResBlock). The EE block aims to strengthen edge information during feature extraction, thereby improving the model's ability to recognize object boundaries. The Multi-ResBlock captures multilevel features in the image through convolutions of different scales, enhancing feature representation. The EE block integrates an ECA Net-based [47] efficient channel attention to enhance edge responses and boundary perception. The Multi-ResBlock consists of three parallel convolution paths: the first path employs a 3×3 convolution, the second path uses a 5×5 convolution, and the third path incorporates a 7×7 convolution. The filter allocation is set as int(w∗0.167), int(w∗0.333), and int(w∗0.5). This design enables the network to simultaneously capture feature information at different scales within the same layer, enhancing its understanding of image details.

In the decoder, the network progressively restores spatial resolution. The enriched features from the encoder are transmitted to the decoder through skip connections, which are modulated by attention gate (AG) blocks. The AG block plays a crucial role by dynamically reweighting incoming feature maps, enabling the network to focus on the most salient regions for segmentation. It takes two inputs—the skip-connection features and a gating signal from a deeper decoder layer—and uses 1×1 convolutions with nonlinear activations to generate an attention map. The attention map is then upsampled to the size of the skip features and fused

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with them, ensuring that task-relevant structures are emphasized and irrelevant responses are suppressed. This design helps maintain high segmentation accuracy and stability under complex and variable inputs. To fully exploit the complementary information across semantic scales, we introduce two MoH modules into the decoder to enable bidirectional feature interaction. Specifically, MoH first uses fine-grained features as the query (Q) to retrieve K/V pairs generated from coarse-grained features, thereby incorporating global category constraints and reducing class confusion in fine-grained predictions. Conversely, it uses coarse-grained features as the query to retrieve K/V pairs from fine-grained features, capturing local detail information and enhancing boundary delineation and detail representation of coarse-grained categories. Unlike conventional multihead attention that uniformly sums all attention heads, MoH treats each head as an expert and employs a learnable weighting strategy to selectively activate the most relevant heads, enabling efficient fusion and precise transmission of cross-scale semantic information.

In the dynamic fusion (DF) block, the original feature X and the MoH-enhanced feature $\tilde{X}$ are jointly fed into the fusion process. The feature $\tilde{X}$ passes through a sigmoid function to generate a fusion coefficient α, which dynamically controls the contribution ratio between the two. In summary, the proposed multiscale feature extraction network effectively addresses the multigranularity representation requirements of hierarchical geographic elements through three key mechanisms.

1) Multi-ResBlocks enable simultaneous capture of features at different spatial scales within individual layers.
2) AG blocks dynamically weight feature maps to emphasize semantically important regions across encoding–decoding pathways.
3) Mixture-of-head (MoH) attention modules facilitate bidirectional cross-scale feature interaction, ensuring coherent information propagation between fine- and coarse-grained semantic levels. This architecture provides robust multiscale feature foundations essential for subsequent hierarchical semantic modeling.

B. Dynamic Hi Loss Function Collaboration

Although the proposed network architecture excels at extracting rich, multiscale visual features, its structure alone does not explicitly encode the underlying geoscientific logic and semantic hierarchies that govern class relationships. For instance, a pixel classified as "river" must logically also belong to the parent category "water." Accordingly, this article proposes an HI Loss collaboration network. It is built on a dynamic HI Loss. During training, it uses a dynamic class interaction matrix to impose implicit consistency constraints. This brings hierarchical consistency into the learning objective and aligns cross-level predictions without adding new feature pathways.

This mechanism effectively enhances the hierarchical consistency of the segmentation results by reducing the number of misclassified pixels and improving the alignment between superclass and subclass predictions, as shown in Fig. 5. This demonstrates that HI not only optimizes feature transfer across

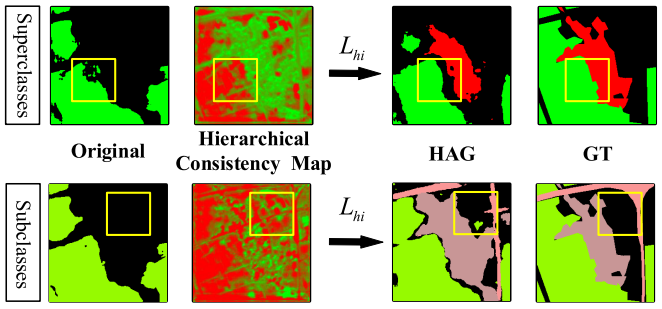


Fig. 5. Hierarchical consistency map with HI Loss effect. Red indicates high-penalty regions where HI Loss applies stronger corrections; green indicates consistent regions.

semantic levels but also strengthens the spatial and categorical relationships within the segmentation framework.

HI Loss is denoted as L_{hi}. This loss function is built based on the standard cross-entropy loss L_{base}. By introducing an additional semantic hierarchy term L_{hier}, it explicitly models the semantic consistency between different classification levels.

This mechanism is formalized through a category interaction matrix $M = \{M_{cf}, M_{fc}\}$, which quantifies the statistical association between coarse-level classes (c) and fine-level classes (f).To prevent the statistics-derived interaction weights from being overly biased by the training data distribution, a learnable calibration matrix is introduced. It is initialized to zeros and updated with a very small magnitude. This enables minor, data-driven refinements while preserving the hierarchical prior. For simplicity, the calibrated interaction matrix is still denoted as M. Among them, w is the associated weight of the category interaction matrix M, which represents the probability that a pixel in the coarse category c belongs to the fine category f. The association weight w_{cf} is computed based on pixel matching statistics. Taking the hierarchical mapping from c to f as an example, the weight w_{cf} is calculated as

$$w_{cf} = \frac{N_{cf}}{\sum_{j=1}^{|F|} N_{cj}} \tag{1}$$

where $|F|$ denotes the total number of fine-grained categories, $\sum_{j=1}^{|F|} N_{cj}$ denotes the total number of pixels assigned to the coarse-grained category c, and N_{cf} represents the number of pixels in the training set that belong to the fine-grained category f under the coarse-grained category c. It is computed by

$$N_{cf} = \sum_{i \in I} \delta\left(y_i^c, c\right) \delta\left(y_i^f, f\right) \tag{2}$$

where y_p^c and y_p^f represent the coarse- and fine-grained labels of pixel p, respectively. The Kronecker delta $\delta(\cdot)$ is defined as 1 when its two arguments are identical, and 0 otherwise.

The semantic hierarchy loss L_{hier} is constructed using the prior hierarchy structure encoded by the matrix M to constrain the model to perceive the hierarchical relationship. The overall HI Loss L_{hi} is a composite objective function that synergizes

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a standard supervised loss L_{base} with a novel hierarchical consistency term L_{hier}

$$L_{\text{hi}} = L_{\text{base}} + \lambda L_{\text{hier}} \tag{3}$$

where λ is a learnable parameter balancing the two terms. The hierarchical term L_{hier} directly penalizes predictions that violate the statistical relationships defined in M. For the coarse-level prediction branch, it is formulated as

$$L^c_{\text{hier}} = \frac{1}{N}\left[\sum_{i=1}^{N}\left(1 - w^i_{f\to c}\right)\right] \tag{4}$$

where $w^i_{f\to c}$ represents the consistency score for pixel i. Crucially, it quantifies the consistency of the coarse-level prediction with respect to the evidence provided by the fine-level prediction. This formulation encourages the model to generate predictions that are consistent across semantic levels. By penalizing inconsistent label pairs (e.g., classifying a pixel as "building" at the coarse level but "river" at the fine level), HI Loss guides the network to learn label assignments that adhere to the predefined geographical semantic hierarchy, effectively reducing cross-level semantic deviations and enhancing the logical coherence of the final segmentation map as shown in Fig. 5.

C. Hierarchical Semantic Prediction Framework

The prediction process begins with an input image I being fed into the HAG Net's encoder to extract a hierarchy of multiscale features. These features are progressively upsampled in the decoder, where skip connections guided by AGs fuse high-resolution spatial details from the encoder with abstract semantic information. At the final decoder stage, the network branches into two parallel prediction heads: one for the coarse-level classification task and one for the fine-level task, producing two probability maps $\hat{\mathbf{y}}_c$ and $\hat{\mathbf{y}}_f$ for the coarse and fine semantic levels, respectively.

During training, the HI Loss is applied to these outputs. The $\mathcal{L}_{\text{base}}$ component computes the cross-entropy loss against the ground truth (GT) labels for each branch independently. Simultaneously, the $\mathcal{L}_{\text{hier}}$ component leverages the category interaction matrix $\mathbf{M}$ to maintain consistency between the predictions $\hat{\mathbf{y}}_c$ and $\hat{\mathbf{y}}_f$. The total loss is the sum of the hierarchical losses from both branches

$$\mathcal{L}^{\text{total}}_{\text{hi}} = \mathcal{L}^c_{\text{hi}} + \mathcal{L}^f_{\text{hi}}. \tag{5}$$

For inference, the final optimal segmentation S^*_I for an image is determined by selecting the label pair $\{l_c, l_f\}$ for each pixel that minimizes the total hierarchical loss, thereby ensuring that the output is the most probable and hierarchically consistent prediction

$$S^*_I = *arg\,min\ \mathcal{L}^{\text{total}}_{\text{hi}}\left(\hat{\mathbf{y}}_i \,\middle|\, \bar{\mathbf{y}}_i, \mathbf{M}, \lambda\right) \quad \forall\, i \in \mathcal{I}. \tag{6}$$

HAG Net reformulates the task as pixel-wise multilabel classification; each pixel is assigned both a coarse-grained (superclass) label and a fine-grained (subclass) label simultaneously. Combined with a dynamic semantic hierarchy constraint mechanism, it enables end-to-end hierarchical semantic segmentation. For feature interaction, the combinations of Q_1 with K_2, V_2, and Q_2 with K_1 and V_1 are adopted to facilitate cross-branch information fusion between fine- and coarse-grained branches. The overall process is illustrated in Algorithm 1.

Algorithm 1 Training of Our Proposed Network

Input: images I, truth label $\bar{y}^c_i$, $\bar{y}^f_i$, the hierarchical interaction matrix M

Output: Model predictions $S^*_I = \{S^*_C, S^*_F\}$

1: initialize network parameters
2: for each training iteration do
3: extract multi-scale features using Multi ResBlocks with depth parameter.
4: enhance encoding features by adding EE block
5: decode with gating signals, W_x and W_g, incorporating Multi ResBlocks.
6: obtain coarse branch feature maps X_C and fine branch feature maps X_F.
7: MoH cross-scale interaction:
path1: Coarse to Fine
- use $Q_1 \leftarrow X_f$ as QUERY
- use $K_2 \leftarrow X_c$ and $V_2 \leftarrow X_c$ as KEY/VALUE
- compute: $F_1 = \delta_{1\times 1}(\text{Attn}(Q_1, K_2, V_2))$

path2: Fine to Coarse use
- $Q_2 \leftarrow X_c$ as QUERY
- use $K_1 \leftarrow X_f$ and $V_1 \leftarrow X_f$ as KEY/VALUE
- compute: $F_2 = \delta_{1\times 1}(\text{Attn}(Q_2, K_1, V_1))$

$\text{Attn}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V$, $\sqrt{d}$ = key/channel dimension per head.
MoH routing: Treat heads as experts, route Top-K heads via shared/routed head scores g_i, balance by α_1, α_2.
8: **Dynamic Fusion (DF)**:
- Fuse: original X_C/X_F feature and MoH enhanced feature $\tilde{X}_C/\tilde{X}_F$
- Adaptive weight $\alpha = \sigma(W_\alpha \tilde{X})$ fusion output $X_{\text{fusion}} = \alpha \times \tilde{X} + (1-\alpha) \times X$

9: generate the coarse-level segmentation prediction S_C and the fine-level prediction S_F
10: compute base losses L^c_{base} and L^f_{base} using cross-entropy
11: dynamically compute hierarchical interaction weights from matrix M
12: compute hierarchical losses: L^c_{hi}, L^f_{hi}
13: compute total hierarchical loss $L^{\text{total}}_{\text{hi}}$
14: generate the final predictions S^*_C and S^*_F based on hierarchical consistency.
15: compute gradients with respect to model parameters
16: **end for**

IV. Experimental Setting and Results

A. Datasets

This study utilizes two representative datasets, the GID datasets and the Ascend Cup dataset, as the foundational datasets for training and evaluating remote sensing semantic

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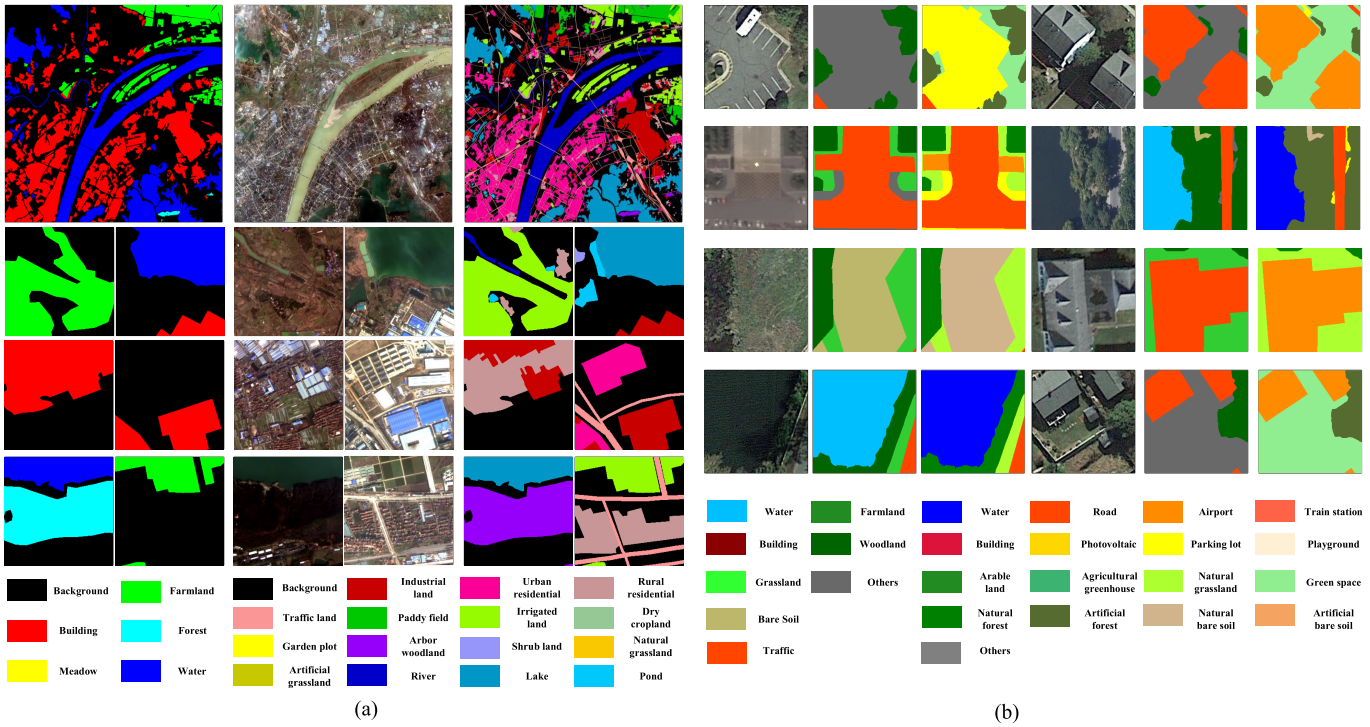


Fig. 6. Data samples taken from GID-5, GID-15 dataset, Ascend Cup2020-8, and Ascend Cup2020-17. (a) Gid dataset. (b) Ascend Cup 2020 dataset.

segmentation models. The GID dataset covers various typical land-cover types, with extensive applications in urban, agricultural, and natural ecological environments. Both GID-5 and GID-15 datasets provide hierarchical classification labels, ranging from coarse-grained to fine-grained categories, effectively enhancing model performance in complex and diverse scenarios. Additionally, the Ascend Cup remote sensing image dataset is a specially designed large-scale competition dataset, comprising millions of high-resolution remote sensing image samples. It encompasses various typical scenes such as Ind.land, urban buildings, and transportation infrastructure. This dataset not only supports the efficient training of deep learning models but also provides a solid foundation for improving model adaptability and generalization capabilities, thanks to its strict annotation standards and diverse image sources.

By integrating these two broadly representative datasets, this study aims to better explore the potential of remote sensing semantic segmentation algorithms in real-world applications, particularly in accurately identifying and differentiating various land-cover types in complex multitarget scenarios. It also seeks to develop models with higher accuracy, robustness, and generalization capabilities to meet the increasingly complex demands of urban and natural environments.

1) GID Dataset:

a) Five Classes: The GID-5 dataset includes a large number of data samples covering various cities in China, consisting of 150 images with a resolution of 6800×7200 pixels, ensuring data diversity for training deep models. However, the dataset contains only five land-cover categories (i.e., built-up areas, farmland, forests, grasslands, and water bodies), which is insufficient for practical applications. The GID-5 dataset is based on high-resolution remote sensing imagery from China's Gaofen-2 satellite. It consists of 150 pixel-level annotated images covering five land-cover types: buildings, farmland, forests, grasslands, and water bodies. The images contain four bands (blue, green, red, and near infrared). In the dataset, red represents buildings, green represents farmland, cyan represents forests, yellow represents grasslands, and blue represents water bodies.

b) 15 Classes: To better meet practical needs, the land-cover categories in GID-5 were further subdivided, resulting in the new GID-15 dataset. By referencing the China Land Use Classification Standard (GB/T21010-2017), a hierarchical classification structure was used to define 15 relevant categories. Specifically, built-up areas were subdivided into industrial land (Ind.land), urban residential land (Urb.land), rural residential land (Rur.resid), and transportation land (Traff.land) to study the detailed distribution of urban areas. Forests were divided into garden land (Garden), arbor forests (Arb.forest), and shrublands (Shr.land) to study vegetation coverage. Grasslands were split into natural grasslands (Nat.mead) and artificial grasslands (Art.mead). Farmland was further subdivided into paddy fields (P.field), irrigated land (Irr.land), and dry land (Drycrop) to analyze agricultural land distribution. Finally, water bodies were categorized into rivers (River), lakes (Lake), and ponds (Pond) for the study of water resources (https://x-ytong.github.io/project/GID.html) as shown in Fig. 6. The semantic association of subclass and class in GID-15 and GID-5 is located in Fig. 7.

2) Ascend Cup 2020 Dataset: This dataset focuses on processing optical remote sensing images with typical land-use classifications. Based on existing remote sensing classification requirements and practical needs, and in reference to geographical national condition monitoring and "Three Adjustments" land classification standards, a feature classi-

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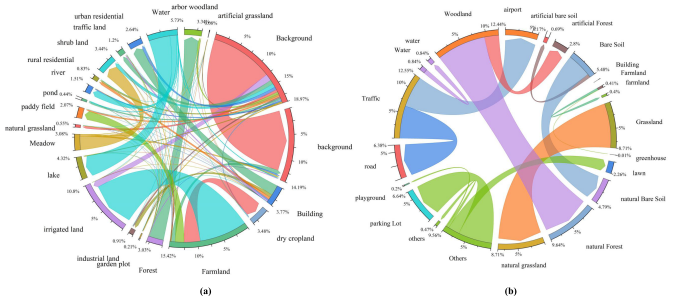


Fig. 7. Semantic association of GID-15 and GID-5, and Ascend Cup2020-17 and Ascend Cup2020-8. The ones that start with a capital letter are the classes in GD-5 and Ascend Cup2020-8, and the others are subclasses. The sector range is the proportion of the category in the total category, and the association line between categories is the association degree of the two classes. (a) GID dataset. (b) Ascend Cup 2020 dataset.

fication system was designed following the "what you see is what you get" principle. The dataset involves eight primary superclasses and 17 secondary subclasses. It should be noted that the primary and secondary categories do not directly correspond to the remote sensing images. In this study, the primary categories used were generated from the secondary subclasses labels as defined by the classification table provided on the official website rsipac.whu.edu.cn/index as shown in Fig. 6. The semantic association of subclass and class in Ascend Cup2020-17 and Ascend Cup2020-8 is located in Fig. 7.

B. Evaluation Metrics and Parameter Setting

To comprehensively evaluate the performance of the proposed model, we utilized several key evaluation metrics: overall accuracy (OA), precision, recall, $F1$ score, and mean intersection over union (mIoU). These metrics have been widely used in previous research and are compared against recognized standards for semantic segmentation evaluation. Specifically, OA measures the proportion of correct predictions among all samples. Precision reflects the proportion of true positives among all samples predicted as positive. Recall indicates the proportion of true positives among all actual positive samples. The $F1$ score is the harmonic mean of precision and recall, used to assess the model's overall performance in classification tasks. The mIoU evaluates the segmentation performance of the model across multiple categories.

C. Results

In this section, we present the performance of the proposed method, evaluated on the GID dataset and the Ascend Cup 2020 dataset, along with a detailed analysis of the results. In Sections IV-C2(a) and IV-C2(b), we comprehensively assess the model's performance on these two datasets, highlighting its superior ability to handle complex scenarios and fine-grained multiclass classification tasks. We use several evaluation metrics to conduct an in-depth analysis of the model's accuracy, generalization ability, and robustness. We also provide a detailed comparison of our method with several traditional and advanced networks based on semantic segmentation experiments on both datasets. The selected comparison networks all employ encoder–decoder structures like ours, enabling effective capture of global contextual information, and include popular transfer learning models. This comparison aims to validate the performance and advantages of our method. Through experimental comparisons with CGGLNet, CMLFormer, and SFFNet on the GID dataset, our results confirm the advantages of the proposed method in terms of overall performance, feature modeling, and semantic segmentation accuracy.

Finally, in Section IV-C, we conduct an ablation study to further validate the effectiveness of HI Loss. By removing or modifying the HI Loss component, we analyze its impact on model performance. The results clearly demonstrate that HI Loss plays a key role in optimizing hierarchical semantic relationships and improving segmentation accuracy. This series of experiments further solidifies the effectiveness and innovation of the proposed approach. Furthermore, in the ablation study, we further validated HAG Net's performance on the 2022 Shandong Land Group Cup Dataset to comprehensively assess its adaptability and effectiveness across scenarios. The results presented above only include categories that actually appear in the test set. Categories with few samples or those not present in the test set are not displayed in the results.

1) Hierarchical Semantic Segmentation in Network Architecture: DeepLabv3+ [25] utilizes the atrous spatial pyramid pooling (ASPP) module to capture multiscale contextual information, effectively enhancing the segmentation model's performance in complex scenes. CGGLNet [47] is a network that guides global context modeling through class information. By estimating the potential class information of pixels, it enhances the aggregation of features of the same class and improves the discrimination of features of different classes. MAE-BG [48] includes an edge detection (ED) branch and a smoothing branch with boundary guidance (BG) capabilities. The ED branch is specifically designed to enhance weak edge features and suppress false responses caused by local textures, implemented using an improved multiattention edge (MAE) detection block. SUNet [49] combines Swin Transformer and UNet architecture for image denoising. It consists of shallow feature extraction, Swin Transformer-based UNet feature extraction, and reconstruction modules, capable of efficiently extracting multiscale deep features and generating denoised images. The gating mechanism of CGGL adaptively captures multiscale local features, performing excellently in complex scenes. Its experimental results on the GID dataset validate its effectiveness in semantic information modeling and segmentation.

2) Hierarchical Semantic Segmentation in Network Architecture: SFFNet [50] utilizes rich spatial information to achieve efficient image segmentation, aiming to tackle the challenges posed by significant grayscale variations in remote sensing images. Its architecture is divided into two stages: the first stage extracts features through spatial methods, capturing spatial details and semantic information; the second stage maps features in both spatial and frequency domains, enhancing the expression of details and over-

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TABLE I
NUMBERS OF PARAMETERS AND FLOPS USED BY DIFFERENT NETWORKS

Network	Params	FLOPs(G)
Deeplabv3+[25]	33,574,280	81.88
CGGL[47]	101,788,595	680.92
MAE-BG[48]	57,526,866	134.37
SUNet[49]	27,147,264	46.73
SFFNet[50]	24,645,428	47.80
CMLF[51]	26,666,382	120.46
HAG	18,714,205	65.98

all features. This multistage design enables SFFNet to perform excellently on the GID dataset, stably handling diversified ground object segmentation tasks. CMLFormer [51] adopts the classic encoder–decoder structure, with the encoder using a lightweight ResNet-18 network to efficiently extract image features while maintaining computational efficiency. The decoder introduces a multiscale local context Transformer block (MLTB), enhancing feature learning ability, capable of capturing image details at multiple scales and improving segmentation accuracy and robustness. The performance of CMLFormer on the GID dataset demonstrates its superiority in handling complex geographical features.

In practical applications, besides accuracy and generalization, model complexity is equally critical. Complexity directly affects the computational cost and efficiency of inference, where parameters (Params) and floating-point operations (FLOPs) are key indicators. Inference time, commonly measured in frames per second (FPS), reflects the model's real-time performance. Table I presents a comparison of parameters, FLOPs, and performance for several models tested on two RTX 3090 GPUs. Generally, dual-classification models tend to be slower due to the need to process two classification tasks. However, the HAG model, with significantly lower Params and FLOPs than complex models like CGGL and MAE-BG, achieves lower inference costs and strikes an excellent balance between efficiency and performance.

a) Results of GID datasets: In this section, we will provide a detailed comparison of the proposed method with some traditional and advanced networks on the semantic segmentation experimental results on the GID dataset. The selected comparison networks all use an encoder–decoder structure, have the ability to capture global contextual information, and encompass currently widely used transfer learning models to verify the performance and advantages of our method. Table II demonstrates the remarkable performance of HAG Net in high-semantic categories, particularly in the building, farmland, and meadow categories. In the building category, a precision of 99.07% and a recall of 98.27% are achieved, outperforming CMLF by 1.51% in precision and 1.08% in recall. In Table III, the $F1$ score of 98.66% and IoU of 97.26% are recorded in the building category, surpassing SUNet by 0.76% in $F1$ score and 1.37% in IoU. Performance gains on high-semantic categories mainly arise from the global constraints and detail enhancement introduced by bidirectional interaction, while

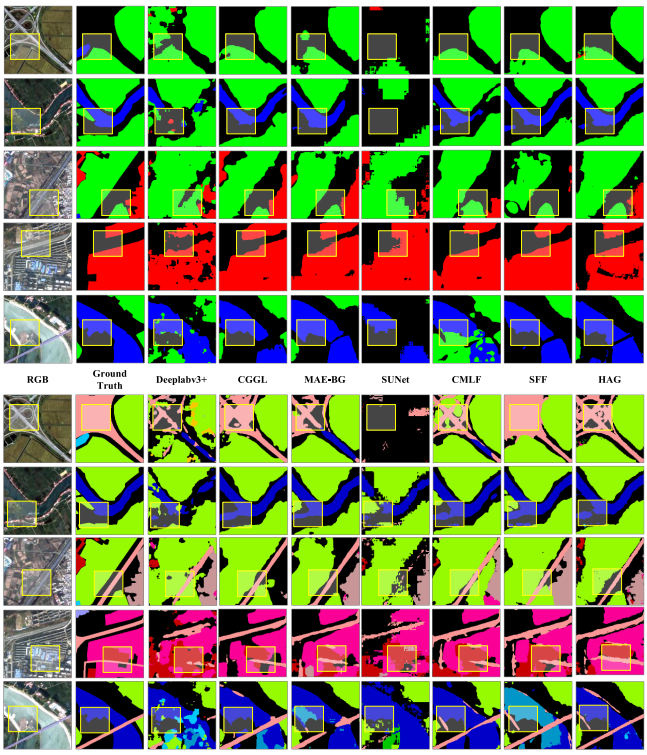


Fig. 8. Performance of traditional and SOTA methods on the GID test set.

the hierarchical consistency constraint reduces cases where a subclass is predicted without its parent, thereby improving stability and separability.

When compared to SFFNet, the IoU in the building category is only 0.77% lower, while the $F1$ score for farmland is 0.88% higher. Tables IV and V further substantiate the advantages in high-semantic categories. In the urban land category, precision reaches 88.21% and IoU stands at 77.9%, outperforming SUNet by 10.92% in precision and 11.19% in IoU. As shown in Fig. 8, the segmentation performance of the networks on the GID test set is compared. DeepLabv3+ utilizes the ASPP module, which is effective in capturing multiscale contextual information. However, its segmentation results tend to be somewhat coarse in regions with intricate details. The CGGL network incorporates class information to guide global context modeling, performing stably but with minor limitations in capturing fine details, particularly in edge regions.

The MAE-BG network enhances segmentation precision through ED and BG, showing strong continuity at object boundaries. While the SUNet integrates the Swin Transformer and UNet architecture for effective context modeling, its performance in complex regions is not as accurate as other networks. The CMLF excels in feature extraction and detail preservation, making it suitable for large-scale segmentation, though there is still room for improvement in edge handling. SFFNet effectively exploits spatial information, demonstrating high performance in complex areas and edge processing, while maintaining excellent segmentation accuracy. Notably, the HAG Net outperforms all other networks, particularly in complex regions and high-semantic categories such as roads and buildings, demonstrating exceptional detail capture and boundary delineation capabilities. Its hierarchical semantic segmentation consistently deliv-

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TABLE II
EVALUATION OF OA AND RECALL RESULTS OF DIFFERENT METHODS IN THE HAG NET ON THE GID-5 TEST SET

Methods	Built-up		Farmland		Forest		Meadow		Water		OA	Recall
	Pre	Recall	Pre	Recall	Pre	Recall	Pre	Recall	Pre	Recall		
Deeplabv3+[25]	98.17	96.41	92.80	93.18	92.13	97.71	96.58	94.48	85.05	83.58	92.19	93.07
CGGL[47]	98.67	98.91	**98.31**	93.51	97.19	94.73	97.17	97.06	80.53	**96.80**	95.03	96.20
MAE-BG[48]	**99.26**	97.26	93.30	95.06	91.81	**99.23**	98.18	95.20	90.03	86.05	93.70	94.56
SUNet[49]	98.67	97.15	94.22	93.39	88.86	84.93	98.43	93.26	83.72	91.67	92.78	92.08
SFFNet[50]	98.78	**99.23**	96.80	94.60	**98.00**	96.55	98.17	**98.78**	87.31	92.68	95.44	96.37
CMLF[51]	97.56	99.19	95.71	92.64	94.24	98.01	96.40	96.54	82.41	87.93	93.29	94.86
HAG	99.07	98.27	96.87	**97.64**	95.90	98.12	**99.08**	96.31	**93.74**	93.08	**96.74**	**96.68**

TABLE III
EVALUATION OF $F1$, MIOU, AND FWIOU RESULTS OF DIFFERENT METHODS ON THE GID-5 TEST SET

Methods	Built-up		Farmland		Forest		Meadow		Water		F1	mIoU	FWIoU
	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU			
Deeplabv3+[25]	97.28	94.71	92.99	86.90	94.84	90.19	95.52	91.43	84.31	72.87	92.99	87.22	85.78
CGGL[47]	98.79	97.61	95.85	92.04	95.95	92.21	97.12	94.39	87.92	78.45	95.13	**95.03**	90.51
MAE-BG[48]	98.25	96.56	94.17	88.98	95.38	91.16	96.67	93.55	87.99	78.56	94.49	89.76	88.36
SUNet[49]	97.90	95.89	93.80	**95.89**	86.85	76.75	95.77	91.89	87.51	77.80	92.37	86.13	86.27
SFFNet[50]	**99.00**	**98.03**	95.69	91.73	**97.27**	94.68	**98.47**	**96.99**	**89.91**	**81.67**	**96.07**	92.62	**91.38**
CMLF[51]	98.37	96.79	94.15	88.95	96.09	92.48	96.47	93.18	85.08	74.04	94.03	89.09	87.61
HAG	98.66	97.26	**96.57**	91.73	96.48	**96.23**	98.17	96.06	87.99	80.76	95.95	92.41	91.09

TABLE IV
EVALUATION OF PRECISION RESULTS OF THE HAG NET ON THE GID-15 TEST SET

Methods	Ind.land	Urb.land	Rur.resid	Traff.land	P.field	Irr.land	Drycrop	Garden	Arb.forest	Shr.land	Nat.mead	River	Lake	Pond	OA
Deeplabv3+[25]	74.05	73.17	82.68	64.28	63.18	78.81	9.17	4.19	87.59	95.90	84.07	63.18	85.59	8.54	78.12
CGGL[47]	69.20	**92.15**	84.73	**82.47**	68.28	88.41	13.62	10.86	97.67	97.86	**95.35**	**85.37**	76.48	10.06	85.24
MAE-BG[48]	75.99	80.62	81.52	73.49	72.92	77.43	**17.26**	1.70	89.35	96.06	86.94	76.13	**86.49**	5.92	79.64
SUNet[49]	61.24	77.29	77.91	23.08	61.96	85.34	1.81	0.52	89.84	96.89	63.28	77.84	77.60	1.33	78.90
SFFNet[50]	**83.31**	79.12	**95.14**	76.63	69.53	90.49	16.88	2.26	**97.86**	96.58	84.73	53.34	72.13	**18.64**	86.43
CMLF[51]	81.13	84.17	84.14	75.76	69.26	90.92	9.71	3.88	96.27	97.68	92.92	69.29	78.93	6.17	84.82
HAG	81.81	88.21	84.08	78.50	**88.70**	**91.21**	15.77	**11.34**	95.63	**98.21**	95.04	70.07	83.20	13.43	**86.94**

TABLE V
EVALUATION OF IOU RESULTS OF DIFFERENT METHODS ON THE GID-15 TEST SET

Methods	Ind.land	Urb.land	Rur.resid	Traff.land	P.field	Irr.land	Drycrop	Garden	Arb.forest	Shr.land	Nat.mead	River	Lake	Pond	mIoU	FWIoU
Deeplabv3+[25]	49.49	63.79	62.12	44.72	38.51	71.65	1.26	3.24	84.39	88.41	58.43	43.28	68.48	6.85	48.90	67.82
CGGL[49]	62.08	**80.49**	**76.34**	**72.64**	**53.79**	81.06	2.87	6.34	**96.36**	**97.15**	61.55	**64.23**	70.86	7.10	**59.49**	77.64
MAE-BG[49]	56.39	70.39	60.89	56.28	40.39	72.05	1.46	1.22	88.10	90.41	58.51	56.58	**72.91**	4.63	52.16	70.49
SUNet[50]	43.96	66.71	50.56	18.07	32.58	75.26	0.59	0.42	84.18	91.89	51.02	47.79	66.96	1.00	45.07	68.18
SFFNet[51]	63.76	74.22	72.31	64.48	47.53	84.11	4.55	1.75	95.79	96.28	65.99	53.34	72.13	**18.64**	58.21	78.25
CMLF[52]	**64.51**	75.64	69.15	65.86	45.82	83.20	3.55	2.79	95.05	96.23	**73.84**	48.34	64.07	4.94	56.64	76.09
HAG	63.30	77.90	74.61	66.14	49.00	**87.53**	**5.43**	**8.63**	94.23	95.66	68.03	55.03	68.42	11.02	58.92	**79.31**

ers high precision across various categories, confirming its superiority as an efficient and stable semantic segmentation method.

b) Results of ascend cup 2020 datasets: HAG Net achieves an accuracy of 78.79% and a recall of 70.83% in the water category, outperforming CMLF by about 4.8% in accuracy and CGGL by 10.3% in Table VI. This indicates that under common disturbances such as dark surfaces, HAG Net leverages bidirectional interaction between classes and subclasses, with the subclass branch enhancing detailed representations to reinforce class-level discrimination, thereby reducing false positives and false negatives and achieving a better precision and recall balance. Looking at the $F1$ score, IoU, and FWIoU analysis in Table VII, HAG Net shines across multiple categories. HAG Net outperforms other methods in both accuracy and IoU across several categories, in Tables VIII and IX. HAG Net achieves an OA of 69.97%, which is 5.6% higher than DeepLabv3+ and 13.2% higher than CGGL.

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TABLE VI

EVALUATION OF OA AND RECALL RESULTS OF DIFFERENT METHODS IN THE HAG NET ON THE ASCEND CUP 2020-8 TEST SET

Methods	Water		Traffic		Farmland		Grassland		Woodland		Bare Soil		Others		**OA**	**Recall**
	Pre	Rec	Pre	Rec	Pre	Rec	Pre	Rec	Pre	Rec	Pre	Rec	Pre	Rec		
Deeplabv3+[25]	68.83	78.63	73.59	78.87	**57.39**	**78.58**	67.15	62.67	82.61	75.45	61.59	68.02	62.33	64.07	71.12	72.33
CGGL[49]	68.53	78.86	78.88	76.83	13.75	57.73	67.00	63.82	83.94	77.21	65.34	58.30	53.74	68.01	71.05	68.68
MAE-BG[49]	69.97	84.40	81.29	79.94	33.65	29.18	65.15	69.30	**85.57**	78.36	63.98	**70.33**	64.77	67.09	73.95	68.37
SUNet[50]	68.23	**87.46**	80.52	75.81	31.18	77.48	66.37	69.26	84.80	78.18	66.50	65.49	59.41	67.04	72.95	57.59
SFFNet[51]	**83.28**	41.52	86.09	78.60	45.42	55.58	**69.96**	58.99	81.13	**79.74**	41.95	68.26	58.19	**71.06**	71.61	64.82
CMLF[52]	83.16	29.31	**86.28**	75.93	47.33	71.28	63.32	60.34	82.01	78.77	42.08	71.31	55.46	68.71	70.25	65.09
HAG	78.79	70.83	82.60	**80.96**	53.45	66.63	68.11	**69.31**	83.66	80.30	**67.30**	68.57	**67.01**	70.94	**75.42**	**72.50**

TABLE VII

EVALUATION OF $F1$, MIOU, AND FWIOU RESULTS OF DIFFERENT METHODS IN THE HAG NET ON THE ASCEND CUP 2020-8 TEST SET

Methods	Water		Traffic		Farmland		Grassland		Woodland		Bare Soil		Others		**F1**	**mIoU**	**FWIoU**
	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU			
Deeplabv3+[25]	73.40	57.98	76.14	61.47	**66.33**	**49.63**	64.84	47.97	78.87	65.11	64.65	47.76	63.19	46.19	69.63	53.73	55.52
CGGL[49]	73.33	57.89	77.84	63.72	22.21	12.49	65.37	48.56	80.44	67.27	61.62	44.53	60.04	42.90	62.98	48.20	55.39
MAE-BG[49]	76.51	61.96	80.61	67.52	31.26	18.52	67.16	50.56	81.81	69.21	67.00	50.38	65.91	49.16	67.18	52.47	59.12
SUNet[50]	**76.66**	**62.15**	78.09	64.06	44.47	28.59	67.79	51.27	81.36	68.58	65.99	49.25	62.99	45.98	68.19	52.84	57.59
SFFNet[51]	55.41	38.32	**82.17**	**69.74**	49.99	33.32	64.01	47.07	80.43	67.27	51.97	35.10	63.98	47.04	63.99	48.27	56.33
CMLF[52]	43.34	27.66	80.77	67.75	56.89	39.75	61.80	44.71	80.36	67.17	52.92	35.98	61.38	44.28	62.49	46.76	54.85
HAG	74.60	59.49	81.77	69.16	59.32	42.16	**68.70**	**52.33**	**81.94**	**69.41**	**67.93**	**51.43**	**68.92**	**52.57**	**71.88**	**56.65**	**60.85**

TABLE VIII

EVALUATION OF PRECISION RESULTS OF DIFFERENT METHODS ON THE ASCEND CUP 2020-17 TEST SET

Methods	Water	Road	Photovoltaic	Parking lot	Garden	Playground	Arable land	Natural grassland	Green space	Natural forest	Artificial forest	Natural bare soil	Others	**OA**	**F1**
Deeplabv3+[25]	69.19	68.31	73.42	63.48	0.00	58.11	68.38	46.63	78.43	55.12	57.69	**71.90**	46.12	66.99	64.37
CGGL[49]	63.51	60.70	69.11	63.91	18.63	14.50	60.50	53.93	78.86	72.96	**61.48**	24.46	56.03	65.01	56.77
MAE-BG[49]	70.34	77.04	79.67	63.21	45.17	31.60	66.03	53.31	77.59	**75.60**	61.37	57.98	68.97	70.09	64.71
SUNet[50]	50.77	62.34	80.34	66.34	14.02	21.50	**73.49**	46.15	75.62	67.06	59.13	41.66	58.40	67.68	59.59
SFFNet[51]	84.52	78.99	87.95	61.55	44.93	**58.45**	69.83	37.83	80.82	60.22	37.43	24.50	59.72	68.60	60.47
CMLF[52]	**86.02**	73.31	**88.83**	55.48	40.96	55.02	60.98	37.11	**81.81**	57.64	33.68	25.83	56.03	65.28	56.55
HAG	77.88	**79.60**	84.00	**68.59**	**45.25**	52.63	71.34	**55.26**	81.62	71.09	61.16	68.91	**70.61**	**73.67**	**69.97**

Different networks exhibit varying performances in complex scenes, as shown in Fig. 9. DeepLabv3+ performs adequately in homogeneous regions but struggles with misclassifications and discontinuities in complex boundaries. CGGL lacks precision in handling fine boundaries, particularly in datasets like the Ascend Cup 2020. MAE-BG performs well in ED and detail retention but requires further improvement in processing small features. SUNet demonstrates stable context modeling but tends to have classification errors in detailed regions.

CMLF performs effectively in large-scale segmentation but struggles in complex boundary processing. SFFNet excels in utilizing spatial information and handling complex areas, but its segmentation accuracy in extremely intricate scenes is slightly weaker than that of HAG Net. HAG Net, on the other hand, achieves superior segmentation results in high-complexity regions, with clear boundaries, detailed retention, and reduced classification errors, validating its reliability and efficiency as a robust semantic segmentation method.

The HAG Net has demonstrated outstanding performance in semantic segmentation tasks on the GID dataset and the Ascend Cup 2020, especially exceling in hierarchical interactive features. Traditional segmentation methods try to preserve semantic information, but single-class label-based networks can usually only assign an independent category to each pixel and struggle to deal with complex overlaps and low separability between categories. Although HAG Net outperforms other networks on most categories, it is still affected by the category see-saw effect induced by multitask learning [52] and struggles to consistently achieve optimal performance on minority classes with limited samples. However, the HAG Net shows a deeper understanding of the structure of ground objects and the ability to optimize segmentation by dynamically

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TABLE IX
EVALUATION OF IOU RESULTS OF DIFFERENT METHODS ON THE ASCEND CUP 2020-17 TEST SET

Methods	Water	Road	Photovoltaic	Parking lot	Garden	Playground	Arable land	Natural grassland	Green space	Natural forest	Artificial forest	Natural bare soil	Others	mIoU	FWIoU
Deeplabv3+[25]	51.32	54.68	60.11	45.35	0.00	**49.97**	47.72	34.85	57.80	43.14	45.05	**53.81**	31.42	44.25	50.47
CGGL[49]	50.53	45.93	58.16	41.61	17.19	13.74	47.49	36.81	58.53	49.53	41.68	22.13	52.39	41.21	48.37
MAE-BG[49]	60.17	60.67	67.00	48.22	26.47	28.80	50.70	34.45	61.38	49.53	47.24	43.91	56.99	48.89	54.39
SUNet[50]	49.26	51.38	64.76	46.19	13.76	18.08	50.32	35.97	58.82	48.98	42.96	37.23	53.98	43.98	51.28
SFFNet[51]	34.38	**63.71**	73.03	49.62	32.00	43.30	47.15	31.17	60.87	48.50	30.28	23.18	43.58	44.68	52.57
CMLF[52]	21.02	60.41	67.28	44.76	26.75	37.33	43.09	30.37	59.82	46.36	28.03	24.66	40.75	40.82	49.22
HAG	**66.64**	61.83	**73.19**	**53.06**	**33.21**	45.01	**54.47**	**41.95**	**65.56**	**54.56**	**48.81**	48.09	**63.11**	**54.58**	**58.60**

modeling the loss function and modeling the category relationship.

At the architectural level, HAG Net couples global semantic information with local geometric and texture details via bidirectional interaction, and at the loss level, it constrains probability mappings between each class and its subclasses with a dynamic class-interaction matrix, thereby reducing error accumulation caused by the downstream propagation of coarse-level errors and the upstream feedback of fine-grained prediction bias. By perceiving and inferring interclass relationships, the HAG Network achieves more efficient and accurate segmentation results in complex remote sensing scenes, providing strong support for the in-depth understanding of the environment and multilevel semantic abstraction. More importantly, the results of the HAG Net at both high and low semantic levels show consistency and mutual promotion. For example, in the Ascend Cup 2020 dataset, the HAG Net performs excellently in segmenting parent classes like "Water" and "Traffic," and its corresponding subclasses "Water" and "Road" also exhibit superior detail capture and classification capabilities.

In contrast, other networks exhibit relatively unstable performance across multiple semantic levels. For instance, as shown in Fig. 9, although the CMLF network can recognize the "Road" category well at the high semantic level, its segmentation effect degrades significantly at the low semantic level due to scene complexity. Similarly, the CGGL network can recognize "Building" in six-category tasks, but recognition failures occur in complex scenes. The robustness of the HAG Network in these aspects further proves its analytical and adaptive capabilities in complex remote sensing scenes, promoting efficient and fine-grained semantic segmentation and demonstrating a deep understanding and analysis ability of environmental structures. At the same time, the HAG Net exhibits excellent generalization performance. Although networks like SFF, CGGL, and CMLF perform well on the GID dataset, their performance declines significantly on the more complex Ascend Cup 2020 dataset. However, the HAG Net maintains excellent performance on the Ascend Cup 2020 dataset, proving its robustness and adaptability when dealing with different data scenes. Even in areas with shadows of street trees that are not labeled in the GT, the HAG Net can still

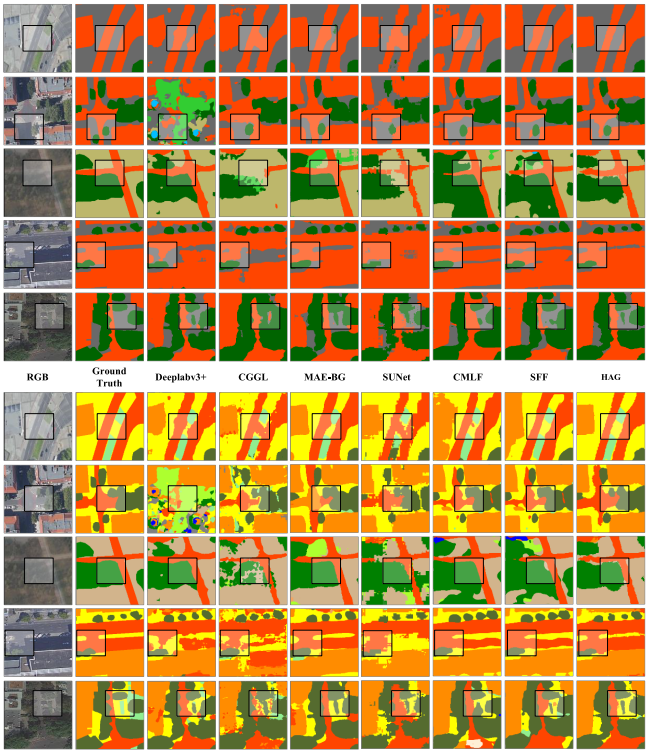


Fig. 9. Performance of traditional and SOTA methods on the Ascend Cup 2020 test set.

achieve classification results with clear boundary localization. This result highlights the unique advantages of the HAG Net in handling complex scenes and detail capture, enabling it to maintain high segmentation performance and precise boundary recognition capabilities in various complex environments.

D. Ablation Study

1) Ablation Study on HI Loss Gains: In the ablation experiments of this article, we compared the performance of the classic DeepLabv3+ network and the dual-branch semantic segmentation network MAE-BG, with and without the addition of the HI Loss function, as shown in Fig. 10. By analyzing the visualizations of our network in remote sensing image

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segmentation, particularly under the influence of the proposed HI Loss, several notable improvements can be observed. First, with regard to farmland areas, traditional segmentation methods often struggle with spectral color changes caused by seasonal variations, making it difficult to accurately segment farmland regions. However, by introducing the HI Loss, association-weighted inference of the relations between farmland and the semantically related fine-grained irrigated land and garden, together with hierarchical consistency constraints, reduces misclassification caused by spectral similarity. Second, after processing by our network, water regions exhibit clearer boundary localization, significantly reducing misclassification between buildings and other categories such as water bodies and grasslands. This improvement is attributed to HAG Net's ability to comprehend and leverage hierarchical relationships and boundary constraints between categories, thereby enhancing the model's capability to handle edge details more effectively.

In the segmentation of fine categories, especially road areas, traditional annotations tend to simplify the task by labeling surrounding regions together as roads, resulting in blurred boundaries during segmentation. With the interaction between HI Loss and fine-class information, our network not only accurately delineates road areas but also effectively distinguishes adjacent nonroad areas, significantly improving the classification accuracy of "irrigated land" and "traffic land." This advantage comes from our network's ability to model category-level dependencies, which allows it to better handle complex types of ground objects and provide more precise semantic segmentation results. Although adding HI Loss to the classic DeepLabv3+ network results in some improvements, the enhancement is relatively smaller compared to other networks when the HI Loss is applied.

2) Ablation of Key Modules in the HAG Network: In the experiments on the GID dataset, we conducted a series of ablation studies on key modules introduced in the HAG Net to assess their impact on overall performance. Specifically, these experiments involved removing or replacing several key modules in the HAG Net, including the multiscale feature extraction module, attention mechanism module, and category relationship modeling module.

The aim was to evaluate their effects on land-cover segmentation accuracy, IoU, and F1 score. By gradually removing these modules, we were able to clearly identify the contribution of each module in capturing contextual information, enhancing semantic consistency, and improving segmentation performance, especially in complex regions. The baseline model was ResUNet, with ED combining a residual block and an ECA module; MRB as the multiscale residual block; AG block as the gated-attention module; MoH as the mixture-of-head attention block; DF as the dynamic fusion block. In the GID-15 test set, the full HAG Net showed improvements of 3.62% in OA, 3.39% in mIoU, and 4.97% in FWIoU as shown in Tables X and XI. These ablation experiments confirm that the effectiveness of the HAG Net is attributed to the synergy of its modules. Each module contributes to improving the network's performance, with the HI module playing a particularly critical role in enhancing segmentation accuracy, especially in complex categories and large-scale datasets. On low-complexity coarse settings (e.g., GID-5), MoH's global interactions can induce negative class transfer, slightly hurting performance. Adding DF mitigates this by adaptively selecting between the MoH branch and the backbone at the pixel level, leading to consistent gains across datasets.

TABLE X
EVALUATION OF THE RESULTS OF HAG IN THE GID-5 TEST SET

Base	EE	MRB	AG	MoH	DF	HI	OA	Recall	F1	mIoU	FWIoU
✓							95.25	95.37	95.21	90.97	91.05
✓	✓						96.14	**96.81**	96.22	92.85	92.59
✓	✓	✓					95.26	96.52	95.95	92.41	91.09
✓	✓	✓	✓				95.48	95.42	95.22	90.99	91.34
✓	✓	✓	✓	✓			95.11	95.74	95.02	90.72	90.76
✓	✓	✓	✓	✓	✓		96.37	96.20	94.87	92.32	93.04
✓	✓	✓	✓	✓	✓	✓	**96.74**	96.68	**96.80**	**93.86**	**93.73**

3) Ablation Study of Semantic Hierarchy-Based Schemes: This section compares three semantic hierarchy-based schemes with our method, denoted as HI1 [53], HI2 [41], and HI3 [21]. All three schemes use tree-structured or coarse-to-fine semantic hierarchies. They adopt a one-way modeling strategy (bottom-up or top-down). They also assume that each fine-grained subclass must belong to a corresponding coarse-grained class.

HI1 [53] adopts a bottom-up modeling strategy. It aggregates predictions of subclasses to their ancestor nodes. It derives high-level probabilities from leaf-node predictions. It then computes cross-entropy losses at multiple levels to constrain the consistency between coarse- and fine-grained predictions. HI2 [41] proposes a recursive coarse-to-fine semantic segmentation framework under image-level supervision. It repeatedly re-estimates and reweights coarse foreground regions by combining category information with iteratively generated pseudo-labels. HI3 [21] corresponds to the HSSN framework. It emphasizes hierarchical heterogeneity constraints. The network is required to predict scores for all nodes in the hierarchy. Explicit constraints are introduced between nodes to ensure that predictions follow the tree structure. HSSN further imposes hard tree-based constraints: 1) once a pixel is predicted as a given coarse category, it must also belong to all of that category's ancestors and 2) if a pixel is not predicted as that coarse category, it must not be assigned to any of its descendants.

In this set of experiments, the baseline model is defined as the network with all bidirectional interaction modules removed. To ensure a fair comparison, all semantic hierarchy-based schemes are implemented under the same fully supervised setting on the GID-5 and GID-15 datasets, where pixel-level GT labels are used as supervision. The original hierarchical modeling strategies and loss formulations of each scheme are kept as unchanged as possible. Therefore, the main differences between the compared models lie only in the adopted hierarchical schemes and their corresponding hierarchical losses. In contrast to HSSN, hierarchical interactions in the proposed HI Loss are realized through a category

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TABLE XI
EVALUATION OF THE RESULTS OF HAG IN THE GID-15 TEST SET

Baseline	M_0	M_1	M_r	M	OA		Recall		F1		mIoU		FWIoU	
					C	F	C	F	C	F	C	F	C	F
✓	✓				95.09	85.61	94.71	66.80	94.86	67.64	90.41	57.54	90.75	78.09
✓		✓			95.78	86.64	96.16	67.08	95.66	66.70	91.79	57.53	91.96	78.28
✓			✓		96.37	86.37	96.23	67.80	95.91	68.61	92.32	58.67	93.05	79.17
✓				✓	**96.74**	**86.94**	**96.68**	**69.00**	**96.80**	**68.81**	**93.86**	**58.92**	**93.73**	**79.31**

TABLE XII
PERFORMANCE EVALUATION OF DIFFERENT HIERARCHICAL MODELING SCHEMES ON THE GID-5 AND GID-15 TEST SETS

Base	EE	MRB	AG	MoH	DF	HI	OA	Recall	F1	mIoU	FWIoU
✓							82.47	64.76	64.75	54.01	73.67
✓	✓						84.24	65.22	65.74	55.53	76.34
✓	✓	✓					83.51	66.26	66.91	56.67	75.67
✓	✓	✓	✓				83.52	64.95	65.62	54.72	74.84
✓	✓	✓	✓	✓			84.25	67.37	67.00	56.29	75.86
✓	✓	✓	✓	✓	✓		86.35	67.82	68.62	58.66	78.09
✓	✓	✓	✓	✓	✓	✓	**86.94**	**69.00**	**68.81**	**58.92**	**79.31**

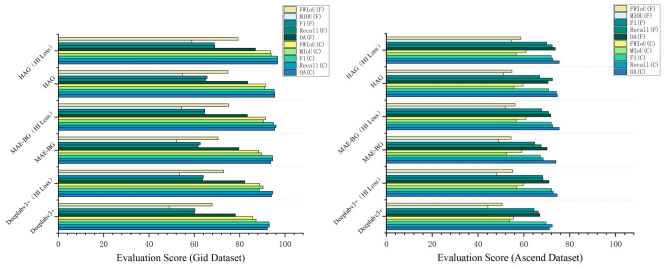


Fig. 10. Accuracy evaluation of gid and ascend cup test sets before and after adding HI Loss (F represents subclasses and C represents classes).

interaction matrix, which models interlevel relations and cross-scale coupling in a soft, bidirectional, pixel-wise manner.

1) When the predicted category of a pixel is consistent with the semantic hierarchy, the corresponding loss term is downweighted.
2) When the prediction violates the semantic hierarchy, the loss term is upweighted. The weights in the category interaction matrix therefore quantify the degree to which each predicted category configuration is consistent with the underlying semantic hierarchy.

The results show that aggregating subclass predictions to ancestor nodes and deriving high-level probabilities from leaf-node predictions cannot fully adapt to the complex multilevel attribute structures in remote sensing imagery. Other semantic hierarchy-based schemes can still achieve good performance on the GID-5 dataset through coarse-to-fine refinement or implicit structural constraints. However, they lack explicit modeling of category–feature interactions. As a result, their performance is clearly limited on the more complex GID-15 dataset, leading to a noticeable performance drop, as reported in Table XII

4) Comparative Ablation of the Category Interaction Matrix: In the experiments on the GID dataset, we further conduct ablation studies on different initialization strategies of the category interaction matrix. Table XIII compares four settings: the all-zero matrix M_0, the all-one matrix M_1, the random interaction matrix M_r, and the category interaction matrix M estimated from the training set. All four settings keep the network architecture and the loss function definition unchanged. Only the initialization of the interaction weights is varied. This protocol evaluates the stability and robustness of the model under different interaction priors.

With the all-zero matrix M_0, interclass interactions are suppressed. The heatmap responses become more concentrated. Strong responses mainly appear on high-contrast structures and visually sharp boundaries. With the all-one matrix M_1, interclass interactions are introduced in a uniform manner. The heatmap responses become more even. Response levels increase across most regions. The response differences between structures are reduced. With the random interaction matrix M_r, the interaction weights lack a stable semantic correspondence. The spatial response pattern depends more on local regions and scene variations. The continuity of response strength and the structural consistency are relatively less stable. With the training-set-estimated matrix M, the heatmaps exhibit a more balanced distribution. Responses are more continuous around salient structures and boundary regions. The above phenomena are reflected in the heatmap comparisons in the misclassification phenomena analysis subsection of the discussion Fig. 11(e)–(h).

Overall, these comparisons indicate that different interaction-matrix initializations change the strength of prior information and its discriminability. The all-zero and all-one matrices provide overly uniform priors. The all-zero case introduces almost no cross-level interaction. The all-one case assigns nearly identical weights to all cross-level pairs. Under the small-magnitude learnable calibration setting, these two initializations are less likely to yield sufficiently informative relation patterns and thus tend to perform worse. The random matrix introduces interaction effects without explicit semantic guidance, yet it still achieves competitive results. This indicates that the proposed learnable interaction mechanism provides sufficient learning space for the optimization process, allowing meaningful interaction patterns to be gradually discovered during training. The training-set-estimated matrix is more consistent with the data distribution. It therefore provides more informative

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TABLE XIII

PERFORMANCE EVALUATION OF DIFFERENT CATEGORY INTERACTION-MATRIX CONFIGURATIONS ON THE GID-5 AND GID-15 TEST SETS

Baseline	HI1[54]	HI2[43]	HI3[21]	HI	OA		Recall		F1		mIoU		FWIoU	
					C	F	C	F	C	F	C	F	C	F
✓					95.48	83.52	95.42	64.95	95.22	65.62	90.99	54.72	91.34	74.84
✓	✓				87.94	78.52	87.38	56.47	84.15	56.84	74.40	42.31	79.55	68.57
✓		✓			91.02	77.12	90.35	64.30	90.32	64.70	71.78	27.18	83.70	66.45
✓			✓		95.25	83.45	96.19	61.42	95.26	62.00	91.15	51.59	92.15	72.52
✓				✓	**96.74**	**86.94**	**96.68**	**69.00**	**96.80**	**68.81**	**93.86**	**58.92**	**93.73**	**79.31**

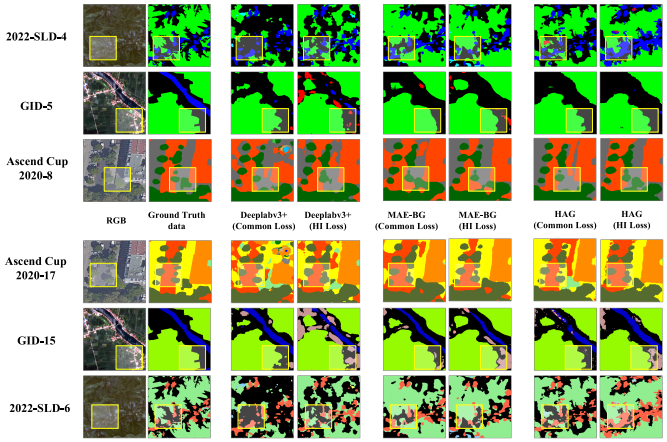


Fig. 11. Effect of key HAG modules and different configuration strategies of the category interaction matrix on segmentation heatmaps. (a) RGB, (b) baseline, (c) baseline with the MoH block introduced, (d) baseline with the DF block introduced, (e) HAG with an all-zero interaction matrix M_0, (f) HAG with an all-one interaction matrix M_1, (g) HAG with a random interaction matrix M_r, and (h) HAG.

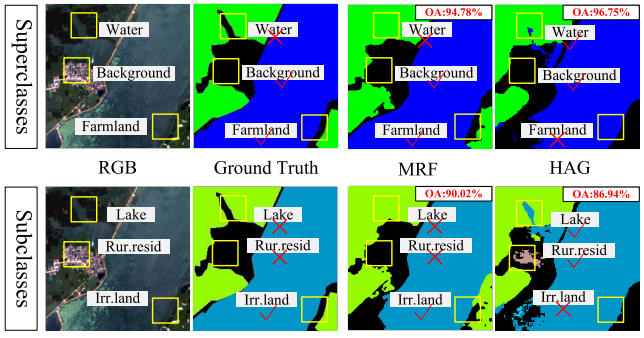


Fig. 12. Presentation on GID and Ascend Cup 2020 and 2022-SLD test sets before and after adding HI Loss.

relation patterns during optimization. With an unchanged loss function definition, these patterns remain generally stable during training. Specifically, the category-level FWIoU varies within 92.37% ± 1.13%, and the subclass-level FWIoU varies within 78.71% ± 0.53%.

V. DISCUSSION

A. Impact of HI Loss Across Different Datasets

To further validate the effectiveness of HI Loss across multiple datasets, we incorporated a new dataset into our experiments: the 2022 Shandong Land Group Cup Dataset (2022-SLD). This dataset is characterized by a relatively small number of categories and was primarily used in this study to evaluate performance. The dataset was divided into two versions: 2022-SLD-6Classes and 2022-SLD-4Classes. The 2022-SLD-6Classes version includes six categories: cultivated land, forest land, grassland, water bodies, urban and industrial/mining land, and a combined category of residential and unused land. This detailed classification approach is designed to rigorously evaluate the performance of HAG Net in multiclass scenarios. In contrast, the 2022-SLD-4Classes version simplifies the classification scheme into four broader categories: water bodies, cultivated land, vegetation, and buildings. As shown in Fig. 12, the 2022-SLD dataset contains some categories that are difficult to distinguish visually or are inaccurately labeled due to annotation tools and human error, particularly with respect to the boundaries and textures of water bodies. These challenging categories showed significant improvement in segmentation performance when using HAG Net. Leveraging the multilevel semantic association mechanism provided by HI Loss, HAG Net effectively captures contextual relationships between categories, significantly enhancing the segmentation of easily confused classes.

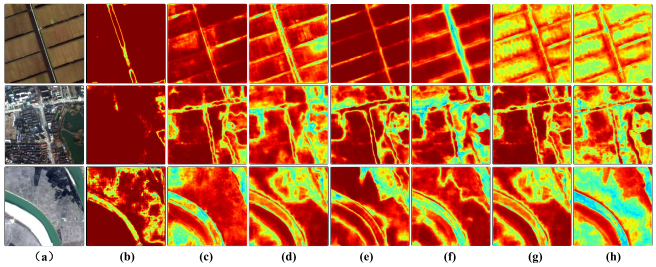


Fig. 13. Comparison between HAG and the traditional multilevel semantic method MRF is presented, where check marks indicate correctly classified pixels, while cross marks denote misclassified pixels.

B. Comparison With the Traditional Hierarchical Semantic Segmentation Method

Given MRF's significant advantage in capturing spatial dependencies between geographic elements, this section compares our approach with the hierarchical semantic also demonstrates robust adaptability and reliability in datasets with fewer categories.

The semantic segmentation approach proposed in [9] addresses the inherent limitation of conventional MRFs,

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namely their reliance on single-granularity information, by establishing a unified framework that jointly leverages multi-granularity cues (pixel- and object-level) together with multiple semantic category levels. As illustrated in Fig. 13, both MRF and HAG exhibit strong consistency in the classification of superclasses and subclasses. MRF, benefiting from postprocessing techniques, demonstrates superior edge preservation. However, due to its strong reliance on labeled data during the classification process, the initial classification serves as the foundation for its final output. Consequently, regions such as rural residential and lake, which are not explicitly marked in the GT, are also absent in MRF's classification results. In contrast, HAG more effectively identifies these regions.

C. Analysis of HAG Impact and Misclassification Phenomena

Heatmap comparisons show that, compared with a baseline without the key modules (MoH, DynamicFusion, and HI), HAG achieves clearer interclass separation and more accurate boundaries in multiclass scenes. Responses in typical regions such as residential areas and farmlands are more stable. Small objects are preserved more completely, with fewer missed detections and less merging. These results indicate that the above modules effectively improve segmentation quality as shown in Fig. 11. As shown in Fig. 12, the introduction of HI Loss in DeepLabv3+ results in only a limited improvement in classification accuracy. This is primarily due to the atrous convolution in ASPP, which captures broader contextual information when processing multiscale features. However, this also leads to excessively dispersed feature aggregation, thereby weakening category distinguishability and reducing the hierarchical constraints imposed by HI Loss. Furthermore, the category transfer matrix in HI Loss relies on the overall distribution pattern of the training set, which some test samples may not fully conform to, resulting in accuracy fluctuations for certain categories. Although HI Loss enhances intercategory associations to some extent, its effectiveness in DeepLabv3+ is constrained by feature extraction inconsistencies and differences in data distribution between the training and test sets, leading to local instability in performance. Compared with other networks, the optimization stability and consistency of DeepLabv3+ after incorporating HI Loss are somewhat inadequate due to its architectural limitations.

This section further examines common misclassification phenomena. As illustrated in Fig. 14, classification errors persist for the "water" category at both semantic levels, primarily for two reasons: 1) the spectral characteristics of traffic roads in the original image resemble those of water bodies and 2) the category interaction matrix establishes an association between roads and water sources, leading to misclassification. Additionally, buildings near traffic roads are frequently misclassified as urban residential areas due to their spatial adjacency. This spatial correlation is particularly pronounced in urban environments, where traffic land adjacent to rural residential areas is sometimes classified as "urban land," especially when these areas share structural similarities such as building density and materials. Furthermore, since urban residential areas appear more frequently in the training set, the model tends to prioritize these categories during learning.

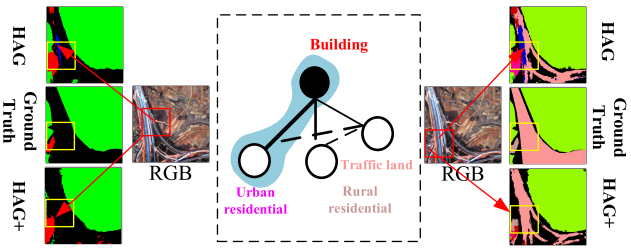


Fig. 14. Typical representations of classification errors. The Conv2D layer in the network and multiplied it with a custom function to achieve adaptive adjustment, and this structure is named HAG+.

Notably, these misclassification patterns validate that the model effectively captures features at both semantic levels and their interdependencies during classification. To address this issue, a Conv2D module was introduced into the network, coupled with a custom function for adaptive adjustments, forming the HAG+ structure. This modification significantly improved the model's stability in recognizing rural residential areas. However, it also led to a slight decline in overall performance, as some farmland regions were not accurately classified. Compared with the original HAG Net, the optimized model exhibited a minor reduction in OA. Therefore, further research is needed to refine the influencing factors and enhance the model's ability to infer spatial relationships more robustly.

VI. Conclusion

The proposed deep semantic segmentation method introduces hierarchical dependency relationships between categories and a dynamic loss adjustment mechanism and demonstrates significant advantages in high-resolution remote sensing image segmentation. This method breaks through the limitations of traditional "flat structure organization" or single-category affiliations by using a special category interaction matrix to capture specific relationships between categories, dynamically integrating them into the loss function to achieve forced constraints and collaborative optimization among multiple categories. This enables the segmentation process not only to have efficient contextual information capture capabilities but also to exhibit extremely high precision and stability in complex backgrounds and multilevel category relationship scenes.

Experimental results show that our method exhibits excellent performance in both coarse and fine category segmentation tasks, proving its outstanding ability in fine-grained classification and complex scene processing. Especially through introducing the dynamic loss adjustment mechanism, our method has reached new heights in category relationship modeling and multiscale feature fusion, effectively reducing segmentation errors and enhancing the clarity of edge processing. This innovative interactive optimization process allows the relationships between class and subclass to influence each

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other during segmentation, improving overall segmentation performance.

Furthermore, ablation experiment results further verify the key role of each module in enhancing segmentation performance, proving the unique contribution of HI Loss in capturing ground object structures and category interaction relationships. Future work will focus on exploring more complex category hierarchical structures and optimizing dynamic loss adjustment strategies to further improve segmentation precision and the model's generalization ability. We also plan to extend this method to more types of remote sensing image data to verify its applicability in different environments and application scenarios. Through continuous improvement, we expect to continue achieving higher segmentation precision and model stability when dealing with more complex and diverse remote sensing tasks. In addition to the three datasets used in this study, the widely available land-use and land-cover (LULC) multisemantic-scale datasets will also be incorporated into the research to further enhance the applicability and research value of this method, allowing it to demonstrate greater potential in more complex scenarios.

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